

CHE 321 – Chemical Reaction Engineering

Fall 2019

Homework Assignment – 2 (40 points)

Due: 12 PM, 7th October 2019

Problem-1

Consider the formation of p-nitrophenol from p-nitrophenyl trimethyl acetate. The process is known as enzymatic hydrolysis and it occurs in the presence of the enzyme elastase. Along with the formation of p-nitrophenol, trimethyl acetic acid is also formed which is an undesired by-product. p-nitrophenol is an important intermediate in the manufacture of several pharmaceuticals. Your role as a Chemical Engineer is to maximize the production of p-nitrophenol. The reactions can be denoted as



where E denotes the enzyme elastase, S denotes the substrate p-nitrophenyl trimethyl acetate, ES denotes enzyme-substrate intermediate and A denotes the trimethyl acetic acid. The rate of the reactions 1 and 2 are given by

$$r_1 = \frac{k_1 C_S}{K_M + C_S} \quad 1$$

$$r_2 = k_2 C_P \quad 2$$

where C_S and C_P denote the concentrations of the substrate and the product. k_1 and k_2 are the rate constants given by 0.015 s^{-1} and 0.0026 s^{-1} . K_M is the Michaelis – Menten constant and is given by 5.53 mol/m^3 . All the reactants and products are in the liquid phase. The initial concentrations of S and E are 0.5 mol/m^3 and 0.001 mol/m^3 .

1. Batch Reactor

Consider the above reaction to occur in a batch reactor for 15 minutes.

- Plot **the concentration profiles of S, P and A as a function of time** in a single figure.
Plot **the selectivity of P with respect to S as a function of time**. (6 points)
- What is the **desired operation time** for the batch reactor to get the **maximum selectivity of P**. (2 points)

CHE 321 – Chemical Reaction Engineering

Fall 2019

2. Plug Flow reactor

Plug flow reactor (PFR) is a continuous ideal tubular reactor where the velocity profile is flat. Consider a PFR of length 5 m and diameter 10 cm. The velocity of the reactants is 0.05 m/s and assume that the **velocity remains constant** throughout the reaction. The reactor is operated in a **steady-state** and assume that the **mass transfer by diffusion is negligible when compared to convection**. Consider **cartesian coordinates** and a **1D** assumption.

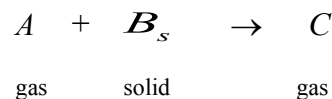
- Derive the **reduced PFR model equation** from **equation 41** of class notes along with the assumptions considered. (3 points)
- Plot the **concentration profiles of S, P and A as a function of the length of the reactor** in a single figure. Plot the **selectivity of P with respect to S as a function of length of the reactor**. (5 points)
- What parameters can be changed to improve the conversion of the reactant S. (2 points)
- Among the batch and plug flow reactors which reactor do you prefer for **efficient operation** for the considered enzymatic hydrolysis and why. (2 points)

Instructions

- Solve the ODEs in the MATLAB using the function **ode 45**.
- Provide the complete code along with the functions for full credits.
- Mention all the considered assumptions.
- Submit the homework in a zip file. All the figures and explanation should be in a **pdf** (*CHE321_HW2_LastName_FirstName.pdf*). The MATLAB codes should be submitted in **“.m files”**. Submit separate scripts and function files separately for both problems. (*prob1_yourinitial.m, prob1func_yourinitial.m, prob2_yourinitial.m and prob2func_yourinitial.m*)

Problem-2

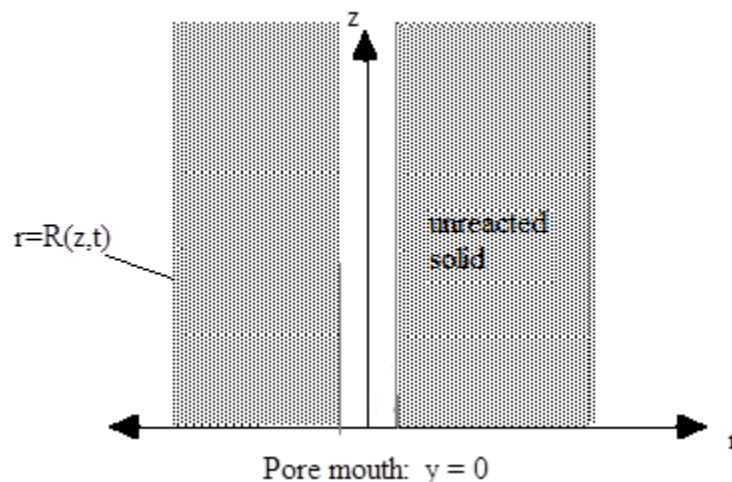
(A) Consider a non-catalytic solid-fluid chemical reaction within **asymmetric** pore of **infinite length** and **circular cross-section of radius initially** R_o occurring at the surface and **the solid is in excess** and after some time the system reaches **steady-state in terms of the concentration**. The consumption of the porous solid is to be modeled via the **analysis of a single pore**. The pore at time t appears as shown in the figure below. The **specific volume is constant**. The **axial diffusion can be neglected**. The System is 2-D (r and z)



The reaction is first order in the concentration of A with an **area-specific rate constant** $\hat{k}$. The concentration of A at the **pore mouth may be assumed to remain constant** at C_{Af} .

CHE 321 – Chemical Reaction Engineering

Fall 2019



- i. Formulate the differential equation for the concentration of gas A neglecting radial diffusion but of course including axial diffusion with coefficient D along the pore. **[5 points]**
- ii. Identify the boundary conditions for the diffusion of the gas within the deforming pore. Neglect external mass transfer resistance. State your assumptions. **[5 points]**
- iii. Obtain the differential equation for $r(z,t)$ representing the shape of the pore assuming C_s to be the molar density of the solid. Provide initial and boundary conditions for $r(z,t)$. **[5 points]**

(B) Consider the combustion of the coal. The air flows at velocity of 1 cm/s.. the diffusivity coefficient of air is $1.6 \times 10^{-5} \text{ m}^2/\text{s}$ and $C_{A,f}$ is 1 mol/m^3 . The concentration of the oxygen is 0.2 mol/m^3 . The reaction constant is $0.00009 \text{ (cm}^2.\text{s)}^{-1}$ at 400°C and mass transfer coefficient of air is $1.5 \times 10^{-3} \text{ cm/s}$. The pore diameter at $t = 0$ is 0.5 cm. and the volume of 13.3 cm^3 coal /mol O_2 is consumed by air. Plot the profile of reaction rate as a function of radius and length in 30 minutes of combustion at length of 1 cm. **[5 points]**

